Answer ALL Questions

Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks

1. a) Use of HVDC is preferred for subsea cables connecting offshore wind farms which are located far away (> 100 km) from the onshore grid Explain why this is so.

[3]

- Cables have larger capacitance per unit length compared to that of an equivalent overhead line.
- Charging current drawn by the cable capacitance increases with cable length (due to larger total capacitance) and operating voltage ($|i_C| = \omega CV$). For long high-voltage cables, charging current could be prohibitively high and limiting the active current carrying capacity of the cable.
- Use of HVDC virtually eliminates the above problem of charging current and thereby, retaining the current (and hence active power) transfer capability of high-voltage long-distance cables.

[1 mark for each point]

(bookwork – interpretation of material from notes)

i) Give an example where an exception to the above was made and state why it was possible in this case.

[2]

- AC cables were used to connect Hornsea One wind farm to the onshore system despite well over 100 km distance between the wind farm and the connection point to the onshore system.
- The above problem of charging current was solved by installing an intermediate reactive compensation station along the subsea cable route to balance the reactive power locally. Construction of such reactive compensation was found to be cost-effective due to shallow waters in this case.

[1 mark for each point]

(bookwork - interpretation of material from notes)

b) The following statements relate to a point-to-point LCC HVDC link. Justify or rectify the statements with proper reasoning:

i) The current order at the rectifier should be maintained at the rated value irrespective of the AC voltage.

[3]

Maintaining rated current order at the rectifier is not desirable under low AC voltage condition as:

Model Answers

- the reactive power demand of the inverter end increases under low AC voltage at the rectifier
- the reactive power supplied by the filters and capacitors decreases with reduced voltage and could trigger voltage instability

Voltage-dependent current order limit (VDCOL) is used to reduce the current order with voltage below a threshold.

[1 mark for each point]

ii) The value of the short circuit ratio is smaller than that of the effective short circuit ratio under the rated condition.

[3]

The short circuit ratio (SCR) and effective short circuit ratio ($ESCR$) are defined as follows:

$$SCR = \frac{SCL}{P_{DC}}$$

$$ESCR = \frac{SCL - Q}{P_{DC}}$$

[1 mark]

Under the rated condition, the filter banks and reactive power support of an LCC HVDC station provide capacitive compensation i.e. positive Q generation which results in a lower value of ESCR.

[1 mark]

In other words, the equivalent impedance as seen from the LCC HVDC converter station increases due to the presence of the capacitive compensation in parallel with the network reactance. Hence, ESCR is less than SCR.

[1 mark]

iii) Design of the DC-side smoothing reactor involves a trade-off between multiple factors.

[3]

A high value of reactance is needed to

- smooth out the direct current ripple
- prevent discontinuous conduction
- filter dc line harmonics
- prevent consequent commutation failure
- limit fault current near rectifier

[1.5 marks]

A low value if preferred to

- avoid resonance at low (non-characteristic) frequencies

Model Answers

[0.5 mark]

- Air core is preferred to avoid magnetic saturation with DC.
- Air core construction leads to a large size.

Hence, partial air core is typically used.
[1 mark]

iv) In order to take one converter out of service for maintenance, it is enough to interrupt the firing pulses to the thyristors of that converter. [3]

Interrupting firing pulse to all the thyristors could result in

- Overvoltage due to sudden drop in current
- Continuous conduction through previously conducting phases at the inverter end placing the AC voltage on the DC line and direct current on the converter transformer
- Hence, the converter bridge needs to be bypassed when firing pulses are blocked to allow part-loaded operation without disrupting the whole DC link

[1 mark for each point]

(bookwork - interpretation of material from notes)

c) Considering the typical values of firing angle (α), extinction advance angle (γ) and overlap angle (μ), show that the reactive power consumed at each end of a point-to-point LCC HVDC link is within 50%-70% of the active power transmitted through the link. [3]

Power factor at each end is given by

$$\cos \emptyset_r = \frac{V_{dr}}{V_{d0r}} = \frac{\cos \alpha + \cos(\alpha + \mu)}{2}$$

$$\cos \emptyset_i = \frac{V_{di}}{V_{d0i}} = \frac{\cos \gamma + \cos(\gamma + \mu)}{2}$$

[1 mark]

Typically,
$15° \leq \alpha$ and $\gamma \leq 18°$
$20° \leq \mu \leq 25°$

Using above expressions, the power factor varies in the range
$0.82 \leq \cos \emptyset \leq 0.89$

[1 mark]

Model Answers

Hence,

$0.51 \leq \frac{Q}{P} = \tan \emptyset \leq 0.69$

which confirms that the reactive power consumed at each end of a point-to-point LCC HVDC link is within 50%-70% of the active power transmitted through the link

[1 mark]

(problem solving – use of standard formulae and interpretation)

d) A 2.85 GW LCC HVDC link interconnects two 275 kV AC systems. The short circuit impedances of the AC systems at the point of connection with the HVDC link are 13.5 Ω and 14.0 Ω. The minimum stipulated effective short circuit ratio (ESCR) at each end is 3.5.

i) Calculate the rated reactive power capacity required at each end as a percentage of the rated active power capacity.

[3]

$S_A = \frac{V^2}{Z_A} = \frac{275^2}{13.5} = 11.85 \text{ GVA}$

$S_B = \frac{V^2}{Z_B} = \frac{275^2}{14} = 11.43 \text{ GVA}$

[1 mark]

$\text{ESCR} = \frac{S - Q_c}{P_{DC}} \rightarrow Q_c = S - ESCR \times P_{DC}$

$\text{ESCR} = 3.5$

$Q_{cA} = S_A - \text{ESCR} \times P_{DC} = 1876.9 \text{ MVAr}$

$Q_{cB} = S_B - \text{ESCR} \times P_{DC} = 1453.6 \text{ MVAr}$

Reactive power capacity required in terms of percentage of rated active power capacity is given by

[1 mark]

$\frac{Q_{cA}}{P_{DC}} \times 100 = 65.85\%$

$\frac{Q_{cB}}{P_{DC}} \times 100 = 51.03\%$

[1 mark]

ii) What is the maximum power that can be transmitted through the HVDC link if the reactive power provision at both ends is limited to 60% of the maximum power transfer?

[1]

Model Answers

$$P_{DC-A} \leq \frac{S_A}{\text{ESCR}+0.6} = 2.89 \text{ GW}$$

$$P_{DC-B} \leq \frac{S_B}{\text{ESCR}+0.6} = 2.79 \text{ GW}$$

[0.5 mark]

So maximum possible rating of the HVDC link is 2.79 GW
[0.5 mark]

iii) How much power can be transmitted if the stipulated ESCR requirement is relaxed to 3.0 with 60% (of rated activepower capacity) reactive power provision at both ends?

[1]

$$\text{ESCR}_1 = 3.0$$

$$P_{DC-A} \leq \frac{S_A}{\text{ESCR}_1+0.6} = 3.29 \text{ GW}$$

$$P_{DC-B} \leq \frac{S_B}{\text{ESCR}_1+0.6} = 3.17 \text{ GW}$$

[0.5 mark]

So maximum possible rating of the HVDC link is 3.17 GW
[0.5 mark]

(problem solving – use of standard formulae, their variants and interpretation)

Model Answers

2. a) A monopole point-to-point LCC HVDC link has one 6-pulse converter at each end. A proportional controller with gain $K_{pr} = K_{pi} = 5000$ is used for controlling the direct current (I_d) for both converters as shown in Figure 2.1 where, V_{dor} and V_{doi} are the no-load ideal direct voltage at rectifier and inverter end, respectively.

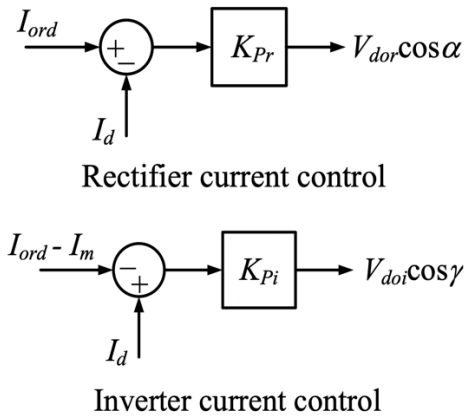


Figure 2.1: Current control scheme for rectifier and inverter

The commutation resistance at both ends is $R_c = 5\ \Omega$ while the DC line resistance is $R_L = 1\ \Omega$. Under normal condition, the AC side voltage at the rectifier and inverter ends are 405 kV and 398 kV, respectively. The turns ratio of the converter transformers at both ends is 0.5 with the low voltage winding on the converter side. The rectifier controls the direct current with a current order $I_{ord} = 1.5$ kA while the inverter operates at a fixed extinction advance angle $\gamma = 18°$. The minimum permissible firing angle for the rectifier is $\alpha_{min} = 5°$. The margin current is $I_m = 250$ A. Calculate the direct current (in kA) through the HVDC link under the following conditions:

i) normal condition. [4]

Rectifier in current control

$V_{dr} = V_{dor}\cos\alpha - R_c I_d = K_p(I_{ord} - I_d) - R_c I_d = K_p I_{ord} - (K_p + R_c) I_d$

[1 mark]

Inverter in constant γ control

$V_{dr} = V_{di} + R_L I_d = V_{doi}\cos\gamma - R_c I_d + R_L I_d = V_{doi}\cos\gamma - (R_c - R_L) I_d$

[1 mark]

Equating V_{dr} from above expressions

$$I_d = \frac{K_p I_{ord} - V_{doi}\cos\gamma}{K_p + R_L}$$

[1 mark]

$$V_{doi} = \frac{3\sqrt{2}}{\pi} E_i T = 268.74 \text{ kV}$$

Model Answers

$I_d = 1.45$ kA
[1 mark]

ii) if the AC voltage at the rectifier end reduces by 4%.

[2]

$$V_{dor} = \frac{3\sqrt{2}}{\pi} E_r T = 268.74 \text{ kV}$$

$$V_{dor}{}' = 0.96 \times V_{dor} = 262.53 \text{ kV}$$

$$\alpha' = \cos^{-1}\left(\frac{V_{dr} + R_c I_d}{V_{dor}{}'}\right) = 11.74°$$

[1 mark]

As $\alpha' > \alpha_{min}$, I_d will be maintained at the same value as in part (i)
[1 mark]

iii) if the AC voltage at the rectifier reduces by 6%.

[4]

$$V_{dor}{}'' = 0.94 \times V_{dor} = 257.06 \text{ kV}$$

$$\alpha'' = \cos^{-1}\left(\frac{V_{dr} + R_c I_d}{V_{dor}{}''}\right) = 0.77°$$

As $\alpha' < \alpha_{min}$, rectifier will be stuck at $\alpha_{min}$and inverter will take over current control
[1 mark]

$V_{dr} = V_{dor}\cos\alpha_{min} - R_c I_d$
[1 mark]

$V_{dr} = V_{di} + R_L I_d = V_{doi}\cos\gamma - (R_c - R_L)I_d$
$= -K_p(I_{ord} - I_m - I_d) - (R_c - R_L)I_d = K_p(I_{ord} - I_m) + (K_p - R_c + R_L)I_d$
[1 mark]

Equating V_{dr} from above expressions

$$I_d{}'' = \frac{K_p(I_{ord} - I_m) + V_{dor}{}'' \cos\alpha_{min}}{K_p + R_L} = 1.3 \text{ kA}$$

[1 mark]

iv) if the AC voltage at the inverter end reduces by 20% with the rectifier end voltage AC voltage remaining the same as in normal condition.

[2]

$$V_{doi}{}' = 0.8 \times V_{doi} = 215 \text{ kV}$$

Model Answers

Rectifier will continue to control current while inverter continues to operate at the same γ
[1 mark]

$$I_d''' = \frac{K_p I_{ord} - V_{doi}' \cos\gamma}{K_p + R_L} = 1.46 \text{ kA}$$

[1 mark]

(problem solving – use of standard formulae, deductions and interpretation of the control modes. This type of problem with finite proportional gain hasn't appeared in the past exams and isn't there in the problem sheet but well-prepared students should be able to make a fair attempt)

b) Two separate AC systems are interconnected by a bipole point-to-point HVDC link using a standard 12-pulse converter arrangement for each pole at both ends. The parameters are given in Table 2.1 below:

Table 2.1: Parameters of the HVDC link

Rated power	$P_{dc-rated}$	1.5 GW
Rated direct voltage	$V_{dc-rated}$	±400 kV
Resistance of each pole	R_L	2.5 Ω
Minimum firing angle	α_{min}	4°
Minimum extinction advance angle	γ_{min}	16°
Commutation resistance per 6-pulse converter	R_c	5 Ω
Current margin	I_m	20%

At a certain point in time, the AC voltages at the rectifier and inverter end are 240 kV and 275 kV, respectively, and the current control is exercised by the inverter. The transformer tap position is set to provide an effective turns ratio $T = 0.63$ with the low voltage winding on the converter side. Under this condition, determine the following using the calculations on a per pole basis:

i) Direct voltage at the rectifier end.

[3]

The calculations are on a per-pole basis

$$B = 2, P_{d-rated} = \frac{1500}{2} = 750 \text{ MW}, V_{d-rated} = 400 \text{ kV}, E_{acr} = 240 \text{ kV}, \text{T} = 0.63$$

$$I_d = \frac{P_{d-rated}}{V_{d-rated}} = 1.875 \text{ kA}$$

[1 mark]

$$V_{dor} = \frac{3\sqrt{2}}{\pi} E_{acr} BT = 408.38 \text{ kV}$$

[1 mark]

$$V_{dr} = V_{dor} \cos\alpha_{min} - BR_c I_d \left(1 - \frac{I_{\text{margin}}}{100}\right) = 392.39 \text{ kV}$$

Model Answers

[1 mark]

ii) Total reactive power consumed by the rectifier. [1]

Power factor angle at rectifier

$$\phi_r = \cos^{-1}\left(\frac{V_{dr}}{V_{dor}}\right) = 16.09°$$

[0.5 mark]

$$Q_r = 2 \times V_{dr} I_d \left(1 - \frac{I_{\text{margin}}}{100}\right) \times \tan\phi_r = 339.52 \text{ MVAr}$$

[0.5 mark]

iii) Direct voltage at the inverter end. [1]

$$V_{di} = V_{dr} - R_L I_d \left(1 - \frac{I_{\text{margin}}}{100}\right) = 388.64 \text{ kV}$$

iv) Extinction advance angle (in degrees) of the inverter. [2]

$$V_{doi} = \frac{3\sqrt{2}}{\pi} E_{aci} BT = 467.94 \text{ kV}$$

[1 mark]

$$\gamma = \cos^{-1}\left[\frac{V_{di} + BR_c I_d \left(1 - \frac{I_{\text{margin}}}{100}\right)}{V_{doi}}\right] = 30.39°$$

[1 mark]

v) Total reactive power consumed by the inverter. [1]

Power factor angle at inverter

$$\phi_i = \cos^{-1}\left(\frac{V_{di}}{V_{doi}}\right) = 33.84°$$

[0.5 mark]

$$Q_i = 2 \times V_{di} I_d \left(1 - \frac{I_{\text{margin}}}{100}\right) \times \tan\phi_i = 781.89 \text{ MVAr}$$

[0.5 mark]

Model Answers

vi) What effective turns ratio T should be set for the transformers at each end to maintain the rated direct voltage at the rectifier end with minimum reactive power consumption at the inverter end.

[5]

$$V'_{dor} = \frac{V_{d-rated} + BR_cI_d\left(1 - \frac{I_{\text{margin}}}{100}\right)}{\cos\alpha_{\min}} = 416.01\ \text{kV}$$

$$T_r = \frac{V'_{dor} \times \pi}{3\sqrt{2} \times E_{acr}B} = 0.642$$

$$V'_{di} = V_{d-rated} - R_LI_d\left(1 - \frac{I_{\text{margin}}}{100}\right) = 396.25\ \text{kV}$$

$$V'_{doi} = \frac{V'_{di} + BR_cI_d\left(1 - \frac{I_{\text{margin}}}{100}\right)}{\cos\gamma_{\min}} = 427.82\ \text{kV}$$

$$T_i = \frac{V'_{doi} \times \pi}{3\sqrt{2} \times E_{aci}B} = 0.576$$

[1 mark for each step]

(problem solving – use of standard formulae, their variants and interpretation)

Model Answers

3. a) Answer the following questions related to the DC-DC converter shown in Figure 3.1. The converter parameters are summarised in Table 3.1:

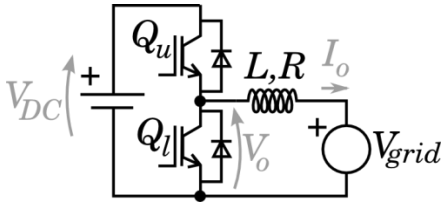


Figure 3.1 – Diagram of a DC-DC converter connected to an ideal DC grid

Parameter	Value	Units	Brief description
V_{grid}	50	V	Nominal DC grid voltage
V_{DC}	100	V	Rated DC link voltage
P_{rated}	500	W	Rated active power delivered to the grid
f_{sw}	20	kHz	Rated switching frequency

Table 3.1 – Characteristic parameters of the DC-DC converter

(i) Write the linear ordinary differential equation (ode) that relates $V_o(t)$, $I_o(t)$ and $V_{grid}(t)$. [2]

Applying Kirchhoff's voltage law going through the loop formed by the two voltage sources and the filter and adding their voltages together, we obtain:

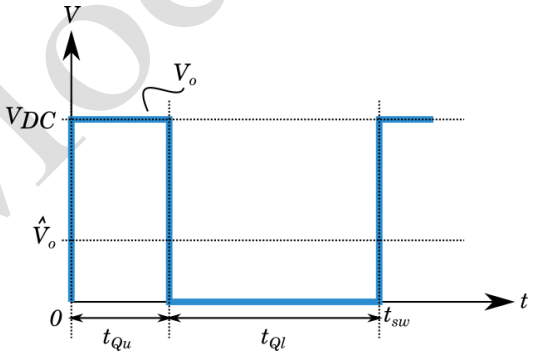
$$V_o(t) - V_{grid}(t) = L\frac{dI_o(t)}{dt} + RI_o(t)$$

[2 marks]

(ii) Q_u and Q_l are controlled using PWM with duty cycle D. Sketch the approximate waveform of V_o for a complete switching cycle and give the relationship between the average of V_o, D and V_{DC}.

[1]

V_o is equal to V_{DC} during the on-time of Q_u and equal to 0 (in both cases, only approximately) during the on-time of Q_l.



The average of V_o is given by $\hat{V}_o = D \cdot V_{DC} = \frac{t_{Qu}}{t_{Qu}+t_{Ql}} \cdot V_{DC}$

[1 mark]

Model Answers

(iii) Calculate the rated DC current and the value of the effective series resistance of the filter, *R*, if we are told that its conduction losses are about 1% of the rated power.

[2]

I_o^{rated}=P_{rated}/V_{grid}= 10 A

[1 mark]

The conduction losses of the filter would be calculated as $P_{loss}=R\,I_o^2$. We are told that these are 1% of 500 W, which gives 5 W. Therefore R=50 mΩ.

[1 mark]

(iv) Calculate the minimum inductance *L* that gives a peak-to-peak current ripple of 0.5 A when the output power is of 5 A.

[4]

This question can be answered using at least two different methods and different levels of simplification. For instance, if *R* is neglected the duty cycle of operation would be 0.5, otherwise the voltage drop over *R* would have to be calculated to find the average of V_o in order to determine *D*.

The on-time of Q_u would be $T_u=D/f_{sw}$=25 µs. During this time, the voltage across the inductor would be V_{DC}-V_{grid}=50 V. The current would travel between its lowest peak and its highest peak during this time, its rate of change would depend on the inductance as:

$$\Delta I_{pk-pk} = T_u \frac{V_L}{L}$$

Therefore, *L* would have to be of 2.5 mH in order for the peak-to-peak ripple to be of 0.5 A.

[4 marks]

b) Answer the following questions related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2.:

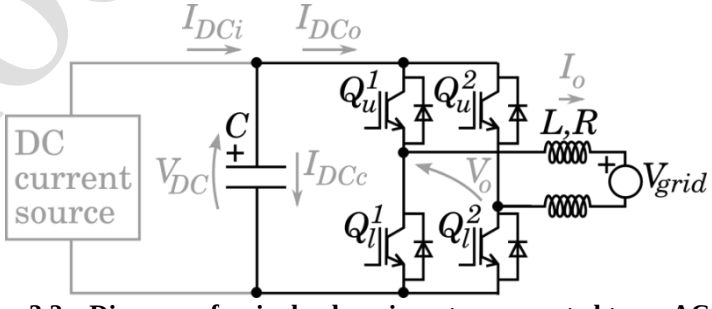


Figure 3.2 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	230	V rms	Rated AC grid voltage
V_{DC}	400	V	Rated DC link voltage
P_{rated}	2	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid

Model Answers

R	0.12	Ω	Equivalent resistance of bridge plus the filter
L	15	mH	Inductance of the filter

Table 3.2 – Diagram of a single phase inverter connected to an AC grid

(i) Calculate the current required to deliver rated power to the grid at unity power factor. Calculate the losses of the filter and its reactive power consumption.

[1]

The current would be: $I_{rated}=P_{rated}/V_{grid}$= 8.7 A. The resistive losses of the filter would be $P_{loss}=RI^2$=9.1 W and its reactive power consumption $Q_{filt}=XI^2=\omega_e LI^2$= 357 VAr

[1 mark]

(ii) Consider an operating point where the reactive power measured at the bridge terminals (ie at V_o) is 0 and the active power is of 1 kW (towards the AC grid) and answer the following questions neglecting any energy losses:

- What is the minimum V_{DC} required by the converter to generate a V_o of 230 V rms? What is the DC component of I_{DCo} at 1kW if V_{DC} is of 400V?

[2]

The bridge can generate a voltage between $+V_{DC}$ and $-V_{DC}$, this range has to be wide enough to reach the peak values of V_o, which are: ±325 V. Therefore V_{DC} must be greater than 325 V.

[1 mark]

The net power extracted from the DC link by the bridge is of 1 kW, therefore the DC component of IDCo is 1kW/400V=2.5A.

[1 mark]

- What is the amplitude of the AC component of I_{DCc} caused by the interaction of the 50 Hz voltage and 50 Hz current? How big should the capacitance of the DC link be for the voltage ripple caused by this current to be less than 2V peak-to-peak?

[3]

The instantaneous power extracted from the DC link by the bridge has a mean equal to 1 kW and a series of AC components due to the interaction between the 50 Hz components and the switching frequency components of voltage and current. Considering the 50 Hz components only, we get:

$$P(t) = v(t) \cdot i(t) = \sqrt{2} \cdot V_{rms} \cdot \cos(\omega_e \cdot t) \cdot \sqrt{2} \cdot I_{rms} \cdot \cos(\omega_e \cdot t + \alpha)$$
$$= \underbrace{V_{rms} \cdot I_{rms} \cdot \cos(\alpha)}_{net\ term} + \underbrace{V_{rms} \cdot I_{rms} \cdot \cos(2\omega_e t + \alpha)}_{100\ Hz\ term}$$

Therefore the amplitude of the 100 Hz term is of $V_{rms}I_{rms}$ = 1kVA. This causes a 100 Hz ripple in I_{DCo} of approximately 2.5 A of amplitude (1.77 A rms).

[2 marks]

Model Answers

Considering that the current from the source, I_{DCi}, is purely DC, all 100 Hz current coming from the bridge must flow through the capacitor of the DC link as I_{DCc}. If we want the voltage caused by this current to cause a voltage ripple of less than 2V peak-to-peak, we need the impedance of the capacitor to be:

$$Z_C \leq \frac{V_{pk-pk}}{I_{pk-pk}} = \frac{2\ V}{5\ A} = 0.4\ \Omega$$

Therefore C must be greater than $(0.4 \cdot 2\omega_e)^{-1}$ F, which gives 3.98 mF.

[1 mark]

(iii) Consider the following closed-loop transfer function of a current controller of the inverter where I_o* is the reference current. Answer the questions listed below:

$$I_o(s) = \underbrace{\frac{1}{\tau s + 1}}_{T(s)} I_o^*(s) - \underbrace{\frac{\tau s}{(Ls + R)(\tau s + 1)}}_{SGd(s)} V_{grid}(s)$$

- Considering τ=0.3 ms, what is the gain and phase delay between I_o*(s) and I_o(s) at 50 Hz?

[3]

This question can be answered by evaluating the frequency response of *T(s)* at 50 Hz. This is done by substituting *s* by *j* and evaluating the function on the corresponding angular frequency:

$$\frac{I_o(j\omega)}{I_o^*(j\omega)} = T(j\omega) = \frac{1}{1 + j\omega\tau} = \frac{1}{\sqrt{1 + \omega^2\tau^2}} e^{-jatan(\omega\tau)}$$

Evaluating for 50 Hz this gives a gain of 0.996 and a phase delay of 5.4º.

[3 marks]

- Determine the proportional gain, K_P, and the integral gain, K_I, of a PI controller that give the closed-loop transfer function, *T(s)*, shown above.

[2]

[BOOKWORK] This closed-loop transfer function can be obtained using an inverse-based design that appears multiple times in the lectures. The controller equation can also be derived by hand if students don't remember the equation:

$$K(s) = \frac{\frac{L}{\tau}s + \frac{R}{\tau}}{s}$$

This can be justified as the open loop transfer function becomes:

$$G(s)K(s) = \frac{1}{Ls + R}\frac{\frac{L}{\tau}s + \frac{R}{\tau}}{s} = \frac{1}{\tau s}$$

which in turn gives the following closed-loop transfer function:

$$T(s) = \frac{G(s)K(s)}{1 + G(s)K(s)} = \frac{1}{\tau s + 1}$$

Model Answers

[2 marks]

c) Answer the following questions related to the AC-DC converter shown in Figure 3.3. The converter parameters are summarised in Table 3.3.:

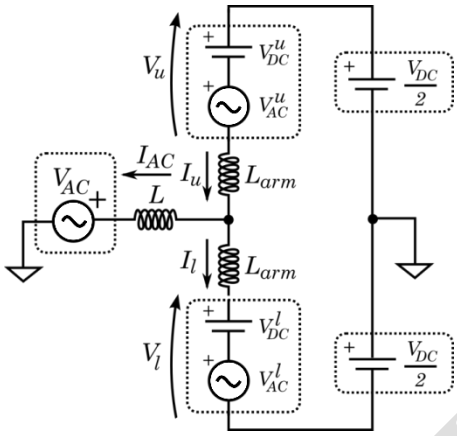


Figure 3.3 – Equivalent circuit of a two-arm single-phase AC/DC modular multilevel converter

Parameter	Value	Description
V_{DC}	1500 V	DC network voltage.
V_{AC}	398 V	AC network voltage (RMS).
f_{AC}	50 Hz	Frequency of the AC network
S_{rated}	300 kVA	Rated power.
X_{arm}	0.11 Ω	Arm reactor inductance.
X_L	0.03 Ω	Phase reactor inductance
R_{arm}, R_L	0	(Negligible) equivalent series resistances

Table 3.3 – Diagram of a single phase inverter connected

(i) Calculate the AC and DC components of V_u, V_l, I_u and I_l when delivering rated power to the AC grid at unity power factor (P_{AC}=300 kW, Q_{AC}=0) under long-term stable operation.

[5]

This question can be solved using the transformed equations of this converter with variables expressed as summative and differential, this makes the process a bit easier and more intuitive; alternatively the equations of the circuit can be solved using the normal variables.

Given P_{AC} and Q_{AC} we can find I_{AC} and the differential voltage the converter must apply, V_{diff}:

$$\bar{I}_{AC} = \frac{300\ kW}{398\ V} = 754\ A$$

$$\bar{V}_{diff}^{AC} = V_{AC} + j\left(X_L + \frac{1}{2}X_{arm}\right)\bar{I}_{AC} = 398 + j64.1\ V$$

On the other hand, the differential DC voltage is zero because no DC current should be fed to the AC grid.

The next step is to solve for the summative variables. Given that no exchange between the upper arm and the lower arm is necessary, we deduce that $I_{sum}^{AC} = 0$. Therefore the

Model Answers

summative AC voltage, $\bar{V}_{sum}^{AC}$, is zero as well. On the other hand even though we don't know the summative DC current yet, because the resistance of the circuit is neglected, we can deduce that $V_{sum}^{DC} = V_{DC}$.

We can find the summative DC current by forcing the balance of power in and out of the converter to be zero given that resistances are neglected and we are told that the converter is under long-term stable operation (the internal energy doesn't drift over time).This gives:

$$V_{DC} I_{sum}^{DC} = P_{AC}$$

Therefore $I_{sum}^{DC} = 200$ A.

Once we've solved for all transformed variables we can undo the transformation to find the real voltages and currents of the two arms:

$$V_u^x = -V_{diff}^x + \frac{1}{2} V_{sum}^x$$
$$V_l^x = V_{diff}^x + \frac{1}{2} V_{sum}^x$$
$$I_u^x = \frac{1}{2} I_{AC}^x + I_{sum}^x$$
$$I_l^x = -\frac{1}{2} I_{AC}^x + I_{sum}^x$$

with *x=AC, DC*.

This gives:

$$V_u^{AC} = -V_{diff}^{AC} = -398 - j64.1 \text{ V}$$
$$V_l^{AC} = V_{diff}^{AC} = 398 + j64.1 \text{ V}$$
$$\bar{I}_u^{AC} = \frac{1}{2} \bar{I}_{AC} = 754 \text{ A}$$
$$\bar{I}_l^{AC} = -\frac{1}{2} \bar{I}_{AC} = -754 \text{ A}$$
$$V_u^{DC} = \frac{1}{2} V_{sum}^{DC} = 750 \text{ V}$$
$$V_l^{DC} = \frac{1}{2} V_{sum}^{DC} = 750 \text{ V}$$
$$I_u^{DC} = I_{sum}^{DC} = 200 \text{ A}$$
$$I_l^{DC} = I_{sum}^{DC} = 200 \text{ A}$$

[5 marks, partial marks awarded for correctly formulating the problem]

(ii) We are told that the currents of the arms of the converter are: $\bar{I}_{AC}^{u}$=42 A, I_{DC}^{u}=35 A, $\bar{I}_{AC}^{l}$=-38 A, $I_{DC}^{l} = 35\ A$. Calculate the AC and DC components of the summative current and the current going to the AC grid. Explain briefly the role of the summative current in the operation of the converter.

[3]

[BOOKWORK] This question requires us to know the definition of the summative and the differential variables:

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$$I_{sum}^{x} = \frac{1}{2} I_u^x + \frac{1}{2} I_l^x$$
$$I_{AC}^{x} = I_u^x - I_l^x$$

for *x=AC, DC*.

Therefore: $I_{sum}^{AC} = 2$ A, $I_{sum}^{DC} = 35$ A, $I_{diff}^{AC} = 80$ A, $I_{diff}^{DC} = 0$ A.

[2 marks]

The summative current is a current that flows between the converter arms and the DC grid without going through the AC grid. The DC component of the summative current is used to transfer power between the DC grid and the converter, which is subsequently transferred to the AC grid using the AC grid AC current. On the other hand the AC component of the summative current is used to exchange power between the upper arms and the lower arms of the converter to balance their energy.

[1 mark]

(iii) Consider the output voltage of one of the arms (ie $V_{AC}{}^{u}$ and $V_{DC}{}^{u}$) to be known and given. What condition must the current of this arm (ie $I_{AC}{}^{u}$ and $I_{DC}{}^{u}$) meet in order to prevent the internal energy of the arm from drifting over time?

[2]

The net power balance across the arm must be zero for the internal energy to not build up over time. This implies that the power the arm exchanges using AC should be the sameas the power exchanged using DC but with opposite sign. Therefore:

$$\Re\{\bar{V}_u^{AC}(\bar{I}_u^{AC})^*\} = -V_u^{DC} I_u^{DC}$$

[2 marks]

Model Answers